

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 8

Status: active execution plan after BA1 minimal identity-ontology closure.

Supersedes: Revision 7 only for forward execution state; R1-R7 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- BA0 responsibility/non-goals: CLOSED by BA0-T3.
- BA1 minimal BAE identity ontology: CLOSED by BA1-T3.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.

2. Closed BA1 identity ontology

BA1 accepts exactly:

BAReferent

BAProposition

BAReferent is the identity family for independently reusable methodology-neutral shared project meaning.

BAProposition is the identity family for independently reviewable methodology-neutral analytical assertions about **BAReferents**.

BAE is an umbrella term, not an additional required metaclass.

A **BAReferent** may denote structural, human/external, informational/resource, behavioral/event, contractual, storage, boundary, state/context or other project meaning. Current evidence does not require dedicated first-class roots for those categories.

3. Closed specialization rule

A new dedicated BAE type/family is justified only if evidence shows both:

1. independent semantic identity across propositions/projections/change; and
2. reusable subtype-specific invariants that cannot be represented honestly as classifications, roles, values or propositions over the closed families.

This prevents future BA phases from turning every useful semantic category into a metaclass by convenience.

4. Phase sequence after BA1

Do not skip phases:

BA2 - Relations and canonical action vocabulary

BA3 - Document-to-BAE derivation, provenance and authority

BA4 - Multi-level human and method projections

BA5 - Controlled vocabulary and optional authoring/extraction assistance

BA6 - Base Analysis closure and regression

Formal analysis-envelope and threat-method work remains after BA6.

5. BA2 objective

BA2 decides the smallest stable methodology-neutral proposition/relation/action structure over the closed BA1 identity ontology.

BA2 must preserve three separations:

semantic identity family
!=
semantic relation / participation role
!=
lexical label / authoring synonym

It must also keep method-owned threat semantics outside Base Analysis.

BA2 may define how reusable semantic classifications are represented, but it may not silently introduce new BA1 first-class identity families. A genuine third-family requirement reopens BA1.

6. BA2-T1 - proposition shape and participation-role lower bound

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

BA2-T1 asks:

What is the minimum methodology-neutral structural shape required for one BAProposition to express reusable project facts over BAREferents without freezing an exhaustive predicate vocabulary or importing a domain/method taxonomy?

Pressure questions

BA2-T1 must test at least:

1. whether binary subject-predicate-object is sufficient or n-ary participation is required;
2. how one proposition can represent roles such as source, destination, information, correlation or responsibility without turning each role into a BA1 type;
3. whether action/behavior identity belongs to BAREferent while action semantics belong to proposition structure;
4. how qualifiers such as condition, state, failure, applicability or realization attach without creating assertion ambiguity;
5. how the same project meaning can participate in different semantic roles across propositions;
6. whether method-neutral classification must be explicit for mechanical projection or can be derived from proposition roles without raw-prose reconstruction.

Guardrails

BA2-T1 must not:

- invent an exhaustive verb list;
- import STRIDE/DFD categories;
- define BA3 provenance fields;
- define BA4 views;
- create ThreatForge implementation classes;
- reopen BA1 unless a concrete identity counterexample is found.

Exit condition

Produce a falsifiable lower-bound proposition structure with explicit alternatives rejected/deferred. Do not yet claim a complete canonical relation/action vocabulary unless the evidence genuinely forces it.

7. Later BA2 work - provisional sequence only

Only after BA2-T1 is reviewed should later BA2 microsteps pressure-test and close the actual relation/action vocabulary.

Likely concerns include:

- stable relation/action semantic identities;
- role cardinalities and n-ary propositions;
- dependency, responsibility, correlation, realization and constraint semantics;
- classification versus relation boundaries;

- lexical canonical labels versus authoring synonyms.

This section is planning only, not semantic acceptance.

8. BA3 - derivation, provenance and authority

After BA2 closure, BA3 must close:

- document-to-BASReferent/BASProposition derivation rules;
- source locators;
- grounded, derived and diagnostic/unresolved materialization;
- identity/equivalence/lifecycle rules;
- acceptance/rejection/stale semantics;
- source-to-analysis and analysis-to-source traceability.

Governed documentation remains project authority.

9. BA4 - projections

Define derived human and method projections from the same accepted BA semantics. No view becomes source authority.

10. BA5 - vocabulary and optional assistance

Define controlled vocabulary and optional assistance only after the semantic contracts exist. Lexical normalization, NLP/LLM proposals and authoring suggestions remain support mechanisms and cannot establish project semantics silently.

11. BA6 - Base Analysis closure and regression

Regress the implemented Base Analysis design against closed corpora and at least one structurally different holdout. BA6, not BA1, closes the complete Base Analysis design.

12. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

The BA0-T2 STRIDE probe remains only a pre-overlay consumer probe.

13. Next authorized microstep

After the BA1-T3 closure package is committed and remotely verified, execute only:

BA2-T1 - minimal BASProposition structural shape and participation-role lower-bound derivation.

Do not start formal STRIDE overlay design, Common Finding schema or implementation work.